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Origin of blood meals in indoor and outdoor resting malaria vectors in western Kenya

A. K. Githeko[a,*], M. W. Service[c], C. M. Mbogo[a,b], F. K. Atieli[a], F. O. Juma[a]

[a]*Kenya Medical Research Institute (KEMRI), Vector Biology and Control Research Centre, P. O. Box 1578, Kisumu, Kenya,* [b] *KEMRI, Kilifi Research Unit, P. O. Box 428, Kilifi, Kenya,* [c]*Division of Parasite and Vector Biology, Liverpool School of Tropical Medicine, Pembroke Place, Liverpool L3 5QA, UK*

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Blood meals were obtained from indoor and outdoor resting malaria vectors in three villages of western Kenya and tested by sandwich ELISA to determine host preferences and their human blood index (HBI). *Anopheles gambiae* s.s. collected indoors at Kisian village had a HBI of 0.97 while that of *Anopheles arabiensis* collected at Ahero was 0.23. However, the HBI of *A. arabiensis* varied depending on the availability of outdoor resting shelters. Most female *A. arabiensis* (98.9%) collected outdoors in granaries at Ahero had fed on cattle. Indoor-collected female *Anopheles funestus* had mainly fed on people (93.0%), but taken at least some of their blood (20.2%) from cattle. Although small numbers of *A. arabiensis* fed on sheep or goats and birds, none of the female *A. gambiae* s.s. and *A. funestus* tested had fed on these hosts. The absence of human-fed *A. arabiensis* in outdoor shelters indicated that exiting after feeding, a behaviour pattern that mitigates indoor insecticidal spraying, is not prevalent in this species in western Kenya.

Key words: *Anopheles arabiensis*; Malaria; Exophily; Blood meal

Introduction

The proportion of infected malaria vectors in a mosquito population depends on, among other factors, the frequency of feeds taken by these vectors on gametocyte-infected people, and species which feed on many host species are less likely to be such good malaria vectors as highly anthropophagic species. The proportion of blood meals taken on humans by vectors such as *Anopheles arabiensis* seems to vary depending on the availability of other hosts, such as cattle.

Important questions in the present studies were whether the numerous malaria vectors resting outdoors had fed on people or animals, and whether freshly blood-fed anophelines caught leaving houses in exit traps had actually fed on people or some other host.

Anopheles arabiensis is generally regarded as being more readily diverted to cattle-feeding than *A. gambiae* s.s., and so in areas having sympatric populations of these two species, such as in the present study area of Kisumu, the HBI of the *A. gambiae*

**Corresponding author. Tel: +254 35–4466. Fax: +254 35–21442.*

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complex can change seasonally, depending on the proportions of the more anthropophagic *A. gambiae* s.s. Since the HBI is a component of vectorial capacity, its value for populations of *A. gambiae* s.l. should theoretically change seasonally depending on the ratio of these two sympatric species.

In several areas of western Kenya trials with permethrin-impregnated bed nets and sisal curtains are being undertaken and so it is important to record baseline data on host preferences of the malaria vectors in order that any changes in feeding behaviour, such as could be induced by impregnated fabrics during, and after, the trials, can be identified.

Materials and methods

Study area

Mosquitoes were collected from Ahero, a village adjacent to a rice irrigation scheme and located 23 km south-east of Kisumu town, from Miwani in the sugar-cane belt some 6 km east of Ahero, and at Kisian, a village 12 km north of Kisumu town. Cattle, sheep, goats, chicken and dogs are kept in all these villages. Homesteads which comprise a number of houses are referred to as Bomas. From May 1989 to May 1990 mosquitoes were collected from three Bomas at Ahero and Kisian and from two Bomas at Miwani.

Sample preparation

Freshly blood-fed *A. gambiae* s.l. and *A. funestus* were obtained from indoor pyrethrum spray collections (PSC) in houses, from searches in granaries and from exit traps fitted to houses. Each blood-fed female had her abdomen squashed on Whatman No. 3 filter paper which was placed in plastic bags and stored at -20 °C until they were tested by the ELISA technique of Service et al. (1986). All samples were labelled according to the method of collection, date, site and house number.

In a concurrent cytogenetic study (Githeko, 1992) all half-gravid samples of *A. gambiae* s.l. collected from Ahero comprised *A. arabiensis*, so it can safely be concluded that all blood-fed adults processed from Ahero were this species while surveys in Miwani showed that *A. gambiae* s.s. comprised 53.5% of the *A. gambiae* complex and *A. arabiensis* 46.5%. To determine the proportions of these two species in Kisian freshly blood-fed individuals were collected from houses and kept in paper cups at 24 °C, until 16.00–17.00 h. By this time they had become half-gravid and the ovaries were dissected out and the half digested blood smeared onto filter paper. The blood smear and the ovaries were labelled with the same number. Ovaries were preserved in Carnoy's fixative and chromosomes prepared according to the method of Hunt (1973). *Anopheles gambiae* s.s. and *A. arabiensis* were identified by the banding pattern on the X-chromosome as described by Coluzzi et al. (1979).

Results

Origins of blood meals of indoor resting populations of *A. gambiae* s.l. *A. gambiae* s.s., *A. arabiensis* and *A. funestus* are as follows.

Anopheles arabiensis

Examples were found of blood-fed *A. arabiensis* caught from houses at Ahero which had fed on each of the host species tested (i.e. human, bovids, goat/sheep and birds) and also presumably a small number of other hosts as indicated by the failure of some blood-smears (5.2%) to be positive for tested hosts (Table 1). A few females had taken blood from more than one host; for example 8.6% of individuals had fed on cattle and either sheep or goats. Large numbers of cattle were present in all three Bomas at Ahero, and by analysing blood meal data for individual Bomas it became obvious that the presence of granaries in some of the Bomas made a great difference to the HBI obtained from indoor resting populations of *A. arabiensis* ($\chi^2 = 38.51$, $p < 0.001$). For instance, in Boma 1 where there were no granaries or other attractive out of door resting sites, females that had fed outdoors on animals sought shelter in houses and this was reflected by the low HBI of 0.23 (Table 2). In contrast in Boma 3 where many cattle-fed *A. arabiensis* entered a granary to rest, the HBI of females caught in houses was 0.67, a value that can be assumed to arise mainly from blood-feds that had fed indoors on people and subsequently remained indoors. It can be argued that host coice of the whole population of *A. arabiensis* are respresented best in collections made from houses without nearby granaries because, irrespective of the origin of the blood meals, most adults rest in houses after feeding because of the paucity of out of door resting sites. Consequently a better estimate of the HBI of *A. arabiensis* in the area is more likely to be 0.23 than 0.67. Despite the availability of granaries in Boma 2, a low HBI (0.27) was recorded and this could have been due to the fact that the sampled house was very close to the animal

TABLE 1

Origins of blood meals of anophelines collected resting indoors and outdoors at Ahero and Miwani

	Hosts					Double feeds		
	Human (H)	Bovid (B)	Goat/Sheep (GS)	Avian (A)	Unknown	H/B	H/GS	B/GS
Indoor resting at Ahero *A. arabiensis*								
Number tested	232	232	186	92	232	232	186	186
Percent + ve	46.98	47.84	15.59	1.09	5.18	3.44	2.15	8.60
Outdoor resting at Ahero *A. arabiensis*								
Number tested	186	186	94	0	0	186	94	94
Percent + ve	0.00	98.92	1.08	0	0	0	0	0
Indoor resting at Ahero *A. funestus*								
Number tested	94	94	0	0	0	94	0	0
Percent + ve	92.55	20.21	0	0	0	12.77	0	0
Indoor resting at Miwani *A. gambiae* s.l.								
Number tested	158	158	60	60	60	158	60	60
Percent + ve	74.05	26.58	0.00	3.33	8.33	3.80	0.00	0.00
Outdoor resting at Miwani *A. gambiae* s.l.								
Number tested	30	30	30	30	30	30	30	30
Percent + ve	3.33	6.67	0.00	0.00	90.00	0	0	0

The double-fed females occur twice in host prefence calculations, consequently the total percentages are >100%.

TABLE 2

Origins of blood meals of *A. arabiensis* at Ahero and *A. gambiae s.l.* at Miwami collected resting indoors in different Bomas

	Hosts					Double feeds		
	Human (H)	Bovid (B)	Goat/Sheep (GS)	Avian (A)	Unknown	H/B	H/GS	B/GS
A. arabiensis at Ahero								
Boma 1, number tested	120	120	120	120	120	120	120	120
Percent + ve	22.50	57.50	16.67	0	3.33	3.33	2.50	7.50
Boma 2, number tested	30	30	30	30	30	30	30	30
Percent + ve	26.67	36.67	13.33	0	23.33	3.33	0	6.67
Boma 3, number tested	110	110	110	110	110	110	110	110
Percent + ve	67.27	26.36	4.55	0.91	0.91	2.73	0.91	4.55
An. gambiae s.l. at Miwani								
Boma 1, number tested	20	20	20	20	20	20	20	20
Percent + ve	95.00	5.00	0.00	0.00	0.00	0.00	0.00	0.00
Boma 2, number tested	74	74	40	40	40	74	40	40
Percent + ve	55.40	50.00	0.00	5.00	12.5	5.40	0.00	0.00

The double-fed females occur twice in host prefence calculations, consequently the total percentages are > 100%.

kraal and as a result several animal-fed mosquitoes flew into the house before encountering a granary.

Anopheles gambiae s.s.

To determine host preferences of *A. gambiae* s.s. at Kisian, blood-fed indoor resting *A. gambiae* s.l. were collected and identified cytogenetically when the ovaries had reached the half-gravid stage, then the remains of the blood meals were tested. Despite the availability of sheep, goats and chickens most (96.6%) of the 87 female *A. gambiae* s.s. collected had obtained blood from people, showing that this species is much more likely to feed on people than *A. arabiensis* at Ahero. Only 3.4% *A. gambiae* s.s. at Kisian had fed on cattle although these were plentiful. Of only four females from Kisian identified as *A. arabiensis* two had fed on cattle and the other two on people.

Anopheles gambiae s.l.

The blood-fed fraction of *A. gambiae* s.l. from Miwani were not chromosomally identified to species before the blood meals were tested, and because of their scarcity during the dry season, blood meal identifications were based on only the period of peak population density, that is from May to July in 1989 and those collected in May in 1990. In 1989 most blood meals taken by *A. gambiae* s.l. at Miwani were from people (74.05%) and only 26.58% were taken on cattle (Table 1); none had fed on sheep or goats although these animals were present in the area, but a small proportion had fed on birds (3.33%), most likely chicken, and this is unusual for any species of the *A. gambiae* complex. Feeding on more than one species of host

was not common and of 158 females tested only six (3.8%) had fed on both people and cattle (Table 1). In Boma 1 in Miwani, where no livestock were kept, 19 out of 20 (95.0%) *A. gambiae* s.l. collected had fed on humans, the other female having fed on a bovid (Table 2) whereas in Boma 2 where cattle were numerous, only 55.4% of *A. gambiae* s.l. females obtained blood meals from people (Table 2).

In May 1990 when the proportion of *A. gambiae* s.s. in half-gravid samples at Miwani was 70% and *A. arabiensis* 30%, 42 out of 43 (97.7%) of the freshly blood-fed females collected indoors had fed on people, and just one had fed on an unidentified host (2.3%). These results show that the HBI for *A. gambiae* s.l. at Miwani, which varied from 0.74 to 0.97, will be influenced by the proportions of *A. gambiae* s.s. and *A. arabiensis* and also by availability of alternative hosts, mainly cattle. The Miwani and Kisian populations of *A. gambiae* s.s. are believed to be genetically homogeneous because they have similar chromosomal inversion frequencies (Githeko, 1992). It would therefore be expected that the blood-feeding behaviour of *A. gambiae* s.s. in Miwani should not be very different from that of Kisian.

There are large-scale sugar plantations at Miwani and the workers are housed in camps and unlike the Bomas at Miwani livestock are absent, consequently these people are the only source of blood meals for *A. gambiae* s.l., and this could increase malaria transmission.

Anopheles funestus

Anopheles funestus females collected resting indoors at Ahero had mainly fed on humans (92.6%) but 20.2% had taken at least part of their blood from cattle of which 12.8% had fed on both people and cattle (Table 1). ELISA tests on these mixed feeds showed weak positive reactions for bovids but strong reactions for humans, possibly indicating that some individuals of *A. funestus* had attempted to feed on cattle but that full blood meals were not obtained, and further feeding was on people. No *A. funestus* had fed on hosts other than humans and cattle (Table 1). The HBI of *A. funestus* females at Ahero was 0.93. Very few *A. funestus* were collected in Miwani and no blood meal tests were carried out.

Origin of blood meals in outdoor-collected vectors

Anopheles arabiensis was by far the most common species collected outdoors in Ahero and Miwani as indicated by chromosomal analysis of the half-gravid fraction of the population. Apart from one *A. arabiensis* that had fed on a sheep or goat (Table 1), all those collected from a variety of granaries at Ahero during different months had fed on cattle. The HBI for all granary-caught *A. arabiensis* was nil. Out of 94 female *A. arabiensis* caught resting in papyrus kiosks, which were specifically built to collect outdoor resting females for capture-mark-release-recapture experiments, 93 (98.9%) individuals had fed on cattle and one on a goat or sheep. These results confirm that the populations collected from these kiosks were similar in their blood-feeding behaviour to those collected from granaries.

The source of blood meal of *A. gambiae* s.l. caught from granaries and unoccupied houses in Miwani is puzzling, because of the 30 smears tested, only one was positive for man and two for cattle; hosts for the other 27 females remain undetermined (Table 1). It is not considered there was a fault in test procedures because all other

samples tested on the same microplate produced acceptable reactions; moreover, the blood-smears were made from freshly blood-fed females and these outdoor samples were stored together with all the other samples. The HBI in outdoor resting *A. gambiae* s.l., of which the majority of half-gravid females (90.2%) were *A. arabiensis*, is therefore 0.03.

Origin of blood meals of A. gambiae s.l. females caught in exit traps

Only 20 freshly blood-fed *A. gambiae* s.l. were captured in exit traps fitted to houses in Miwani, but 70% had fed on people and 5% on cattle. Hosts for the remaining 25% of the females could not be identified, but they were not sheep, goats or birds.

Discussion

Correct identification of hosts of malaria vectors is of paramount importance in epidemiological studies (Service, 1986, 1989), as is the determination of the frequency at which a vector population feeds on humans as this is the true measure of human-vector contact.

Examples were found of female *A. arabiensis* from Ahero which had fed on each of the hosts tested but the predominant hosts were cattle, as reflected by the low HBI (0.23). This, however, does not mean that *A. arabiensis* cannot be an important malaria vector, because its large population size means that the actual numbers feeding on people can be quite large. The *Plasmodium falciparum* sporozoite rate (ELISA) of indoor resting female *A. arabiensis* was about 1.1% (Githeko et al., 1993).

One of the most important findings of this study was that none of the outdoor collected *A. arabiensis* at Ahero had fed on man. This, together with the observation that only 1.1% of females leaving houses during the night of feeding were freshly blood-fed (Githeko, 1992) strongly suggests that exiting from houses after feeding on people is not prevalent in *A. arabiensis* at Ahero. If this is the case then in the absence of insecticide resistance residual house-spraying should be effective against this population at Ahero. There are, however, large numbers of cattle-fed female *A. arabiensis* resting outdoors and it might be thought that these individuals would escape house-spraying but capture-mark-release-recapture studies have demonstrated that although females are usually exophilic, they nevertheless sometimes rest in houses, and consequently come into contact with insecticides (Githeko, 1992). Nevertheless, under intense insecticide pressure it is possible that endophilic and anthropophagic individuals might be greatly reduced or even eliminated to leave genetically zoophagic and exophilic adults. The history of vector control in Madagascar suggests this process can occur. For instance, Chauvet and Rajaonarivelo (1973) observed that after vector control in the Antananarivo area of Madagascar with DDT and dieldrin from 1949 to 1960, the endophilic and anthropophagic populations of *A. arabiensis* and *A. funestus* were eliminated and the residual populations were completely zoophagic and exophilic. However, they hypothesised that there could be a reconstitution of the original populations at a later stage leading to renewed malaria transmission and possible epidemics. In fact in 1986 and 1987 malaria epidemics occurred in the Plateaux of Madagascar and

A. arabiensis was the main vector (Fontenille et al., 1990). In the Taita Taveta area of south-west Kenya house-spraying virtually eliminated *A. funestus*, and reduced the indoor-resting *A. arabiensis* by about 80%, but the remaining 20% were all cattle-fed and no infected vectors could be found, although there was low grade malaria transmission, as observed in young infants (Smith and Draper, 1959).

At irrigation schemes similar to Ahero any increase in rice acreage could mean a reduction in grazing land and reduced livestock and this could cause more *A. arabiensis* to feed on people, and favour increased malaria transmission. On the other hand increased biting could force more people to use bed-nets which might lead to a reduction in disease transmission. Olson et al. (1979) showed in California, that as the population size of *Culex tarsalis* increased, the incidence of St. Louis encephalitis (SLE) and western equine encephalomyelitis (WEE) suddenly decreased, apparently because higher biting densities promoted increased defensive behaviour amongst birds, the reservoir hosts of these viruses. In Guyana a reduction of livestock caused the normally zoophagic *A. aquasalis* to feed on man and as a repercussion malaria returned to the Demerara river estuary some 16 years after it had been controlled (Giglioli, 1963). Another example of the effect of reducing livestock on malaria transmission comes from Indonesia where farmers reduced the number of water buffaloes, thus forcing the normally zoophagic *Anopheles aconitus* to feed on people, and as a result the number of malaria cases increased 5-fold (Service, 1991).

Highton et al. (1979) reported an HBI of 0.39 for *A. arabiensis* resting in houses in the Kano Plains near the Ahero irrigation scheme, but lower HBI's from mosquitoes, collected from homesteads having large numbers of cattle. Similary west of Kisumu town the HBI of indoor-resting *A. arabiensis* is higher than for the Ahero population. Joshi et al. (1975), recorded 92.4% of *A. arabiensis* had fed on people and in subsequent work in the same area Service et al. (1978) found that 85.9% had fed on people. A lower proportion of *A. arabiensis* females (60.0%) from the Kisian and Saradidi areas was found to have taken human blood meals (Petrarca et al., 1991), illustrating that feeding pattern can vary considerably over relatively small areas, and probably also change temporally. In contrast all data for outdoor-resting *A. arabiensis* from the Kisumu area consistently show that about 90.0% of the females feed on bovids (Joshi et al., 1975; Service et al., 1978; Petrarca et al., 1991).

In the Kaduna area of northern Nigeria, all *A. arabiensis* caught in Hausa villages where there were few alternative hosts had fed on humans, whereas in Fulani villages where cattle were more numerous than people 39% had fed on cattle (White and Rosen, 1973). At Mwea in Central Kenya where cattle are numerous the human blood index (HBI) of *A. arabiensis* was reported as only 0.28 (Mukiama and Mwangi, 1990) or even as low as 0.10 (Ijumba et al., 1990). An extreme form of zoophagy in *A. arabiensis* was reported by Randrianasolo and Coluzzi (1987) in a population in the Antananarivo area of Madagascar, where none of the females had fed on people. In fact *A. arabiensis* can exist in the complete absence of man, as was observed in Manyara Game Park in Tanzania (White, 1974). In contrast *A. arabiensis* females were found feeding entirely on man in the Gambela area of Ethiopia, where neither cattle nor any other domestic animals were kept (Krafsur, 1971).

Livestock may provide some degree of malaria zooprophylaxis such as Ahero where the vector is *A. arabiensis*, a species which readily feeds on cattle, but

there would be no protection against highly anthropophagic mosquitoes, such as *A. funestus*. The interactions between livestock, mosquitoes and disease transmission is, however, complex. For example, it has been pointed out that increasing livestock could increase vector populations and thus increase the absolute numbers of mosquitos biting people (Sota and Mogi, 1989; Saul, 1990).

In our study most (96.6%) *A. gambiae* s.s. caught resting indoors at Kisian had fed on humans, and Petrarca et al. (1991) obtained almost identical results. About 15 km west of Kisian the HBI of this species has been found as high as 0.93 (Joshi et al., 1975) and a little later slightly lower HBI (0.80) (Service et al., 1978). Apart from situations where people are outnumbered by cattle, for example in the Kaduna area in northern Nigeria where the HBI of *A. gambiae* s.s. was 0.28 (Service, 1970) and 0.50 (White and Rosen, 1973), the HBI of this species is usually high (0.8–0.9) occasionally even reaching unity (White, 1974; Garrett-Jones et al., 1980, for review).

Although most *A. gambiae* s.l. in Miwani obtained blood meals from people, the HBI is likely to vary seasonally depending on the proportions of *A. gambiae* s.s. and *A. arabiensis* comprising the vector population. ELISA tests have shown that 8.3%, 20.0% and 90% of *A.gambiae* s.l. collected indoors, in exit traps, and outdoors, respectively had fed on hosts other than humans, the exophilic adults were probably mostly *A. arabiensis*. Although only 20 blood-fed *A. gambiae* s.l. were caught in exit traps at Miwani 70% of them had fed on people and were leaving houses soon afterwards. But as there was only one human-fed out of 30 females caught from granaries and unoccupied houses, it is possible that those caught in exit traps would, in the absence of traps, have entered other houses, and not rested outside.

Most *A. funestus* collected indoors at Ahero had fed on people, a feature commonly observed elsewhere in the Afrotropical region. Nevertheless, somewhat surprisingly as many as 20.2% of the females had taken at least some of their blood from cattle and 12.8% had fed on both cattle and humans. In some studies in Uganda and Burkina Faso it has been reported that less than 50% of *A. funestus* fed on people (Gillies and De Meillon, 1968). Recently Fontenille et al. (1990) found that although very few *A. funestus* collected indoors in the Manarintsoa area of Madagascar had fed on animals, 65.0% of the exophilic population had bovid blood. Nevertheless, *A. funestus*, as shown by Pull and Grab (1974) and Beier et al. (1988) is highly anthropophagic and an efficient malaria vector.

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